

Nano Screen Transitions — Motion Timesheet

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Overview

Every page change in Nano is one **push**: the incoming page slides in a full screen width while fading in, and the outgoing page slides out the other way while fading out. Each incoming page then plays its own short entrance on top. Two flows are covered.

Flow	Order	Push spring	Settles in
Kid onboarding	Splash → Email → OTP → Skin → Avatar → Interests → Done	P1: response 0.50 s, damping ratio 0.88	~420 ms
Parent	Child → Email (+ confirm sheet) → Intro → Rules → Address → Invite	P2: stiffness 320, damping 28	~330 ms

- **Stage:** every screen is laid out at 375×812 pt and scaled to fill the device. All offsets here are in those stage points; multiply by the stage scale on device.
- **Exceptions:** Splash → Email is a crossfade with no slide. The confirm-email sheet is an overlay, not a page (see the Bottom Sheet spec).
- **Not covered:** gestures inside a step (skin cycling, OTP digits, avatar swiping), the button press (see the Button v3 handoff).

Motion timesheet

$t = 0$ is the moment the page change starts. W = stage width (375pt). Springs are named in the next section (P1, P2, E1...). "Push" = incoming $\text{translateX} + W \rightarrow 0$ and opacity $0 \rightarrow 1$; outgoing $\text{translateX} 0 \rightarrow -W$ and opacity $1 \rightarrow 0$, both on the same spring. Going back mirrors the directions.

Kid onboarding

Transition	Trigger	Element	Property: from → to	Start	Duration	Curve
Splash → Email	Video ends (~1.93 s) or tap to skip	Splash (video paused on last frame)	opacity 1 → 0	0	~420 ms	P1
		Email page	opacity 0 → 1 (no slide; its backdrop is a still of the video's last frame)	0	~420 ms	P1
		Email card	translateY +40 → 0, opacity 0 → 1	80 ms	~465 ms	E1
		Email CTA bar	translateY +120 → 0	80 ms	~465 ms	E1
Email → OTP	Send OTP or Return	Email page	opacity 1 → 0 (no slide)	0	~420 ms	P1
		OTP page	push in	0	~420 ms	P1
		Keypad	field focuses, keyboard rises	680 ms	system	system
		OTP art	width 500 → 290pt (height = width × 134/290) as the keyboard comes up	with keyboard	~390 ms	E5
OTP → Skin	4th correct digit: keyboard drops, success haptic + toast	OTP / Skin	push	950 ms after the 4th digit	~420 ms	P1
		Skin header	translateY -14 → 0, opacity 0 → 1	+350 ms after push start	~545 ms	E4
		Skin cards (deal-in)	scale 0.9 → slot, translateY +34 → slot, opacity 0 → 1	+350 ms, then +0 / 120 / 240 / 360 ms per card	~545 ms each	E4
Skin → Avatar	Continue	Skin / Avatar	push	0	~420 ms	P1
		Avatar content column	translateY +18 → 0, opacity 0 → 1 (backdrop moves only with the push)	0	~400 ms	E2

Transition	Trigger	Element	Property: from → to	Start	Duration	Curve
Avatar → Interests	Continue	Avatar / Interests	push	0	~420 ms	P1
		Interests content column	translateY +18 → 0, opacity 0 → 1	0	~400 ms	E2
Interests → Done	Continue (3+ picked) or Skip for now	Interests / Done	push	0	~420 ms	P1
		Done icon + text	scale 0.9 → 1, opacity 0 → 1 (CTA bar moves only with the push)	0	~555 ms, 6% overshoot	E3
Done → Splash	Run it again	Done page	push out, backward	0	~420 ms	P1
		Splash	opacity 0 → 1	0	~420 ms	P1

Parent flow

The progress bar (72 × 4pt capsule, y 73) sits above the pages and does not slide; only its fill springs to the next value on P2. Fill per page: Child 13.3/72, Email 2/6, Intro 3/6, Rules 4/6, Address 5/6, Invite hidden.

Transition	Trigger	Element	Property: from → to	Start	Duration	Curve
Child → Email	Continue	Keyboard	drops	0	system	system
		Celebration Lottie	mounts full-screen, plays once (90 frames at 60 fps; fades itself)	0	1.5 s	Lottie
		Child / Email	push	1267 ms (Lottie frame 76)	~330 ms	P2
		Progress fill	13.3/72 → 2/6	1267 ms	~330 ms	P2
		Email field	auto-focus, keyboard rises, action row rides it	+350 ms after push start	system	system

Transition	Trigger	Element	Property: from → to	Start	Duration	Curve
Email → confirm sheet	Continue (valid email)	Sheet	see the Bottom Sheet spec	0	~330 ms	springLight
Email (+ sheet) → Intro	Yes, use this email	Email page (sheet still on it) / Intro	push	0	~330 ms	P2
		Progress fill + colour	2/6 → 3/6; white → purple	0	~330 ms; 350 ms	P2; ease-in-out
		Intro stars backdrop	scale 1.12 → 1 and rotate -6° → 0 (anchor 50%, 30%), opacity 0 → 1	0	700 ms	ease-out
		Intro wallet card	translateY -260 → 201.8; rotateX 24° → 0 (anchor top, perspective 0.45)	80 ms	~730 ms, 8% overshoot (intended swing)	E6
		Intro copy	translateY +12 → 0, opacity 0 → 1	350 ms	350 ms	ease-out
Intro → Rules	Auto after 1.7 s, or tap anywhere	Intro / Rules	push	1700 ms	~330 ms	P2
		Progress fill + colour	3/6 → 4/6; purple → white	1700 ms	~330 ms; 350 ms	P2; ease-in-out
Rules → Address	Continue	Rules / Address	push	0	~330 ms	P2
		Progress fill	4/6 → 5/6	0	~330 ms	P2
Address → Invite	Continue (1+ address)	Let's Go Lottie	opacity 0 → 1, plays once	0	150 ms fade	ease-out
		Address / Invite	push (Address keeps the Lottie's last frame)	when the Lottie ends	~330 ms	P2
		Let's Go Lottie	unmounts	+450 ms after push start	—	—
Invite → Child	Close	Invite / Child	push, backward	0	~330 ms	P2
		Progress bar	reappears at 13.3/72	0	~330 ms	P2

Transition	Trigger	Element	Property: from → to	Start	Duration	Curve
Any page → previous	Back	both pages	push, backward (Back from Rules skips Intro, lands on Email)	0	~330 ms	P2

Pages each play their header Lottie once on appear (Email: `parent_email`, Rules: `parent_shield`, Address: `parent_pin`).

Curves and springs

All springs have mass 1. Use the Reanimated column as is: `withSpring(v, { mass: 1, stiffness, damping })`. Settle = time to within 1% of the target.

Name	Used for	iOS (as written)	Stiffness	Damping	Damping ratio	Overshoot	Settles in
P1	Onboarding page push	<code>.spring(response: 0.5, dampingFraction: 0.88)</code>	157.91	22.12	0.88	0.3%	~420 ms
P2	Parent page push, progress fill	<code>.interpolatingSpring(stiffness: 320, damping: 28)</code>	320	28	0.78	1.9%	~330 ms
E1	Email card + CTA entrance	<code>.spring(response: 0.52, dampingFraction: 0.82)</code>	146.0	19.82	0.82	1.1%	~465 ms
E2	Avatar, Interests entrance	<code>.spring(response: 0.46, dampingFraction: 0.84)</code>	186.6	22.95	0.84	0.8%	~400 ms
E3	Done entrance	<code>.spring(response: 0.5, dampingFraction: 0.66)</code>	157.91	16.59	0.66	6.3%	~555 ms
E4	Skin deal-in, header	<code>.spring(response: 0.55, dampingFraction: 0.74)</code>	130.5	16.91	0.74	3.2%	~545 ms
E5	OTP art shrink	<code>.spring(response: 0.46, dampingFraction: 0.86)</code>	186.6	23.49	0.86	0.5%	~390 ms
E6	Intro wallet card swing	<code>.spring(response: 0.62, dampingFraction: 0.62)</code>	102.7	12.57	0.62	8.4%	~730 ms
springLight	Confirm sheet	<code>.interpolatingSpring(stiffness: 300, damping: 28)</code>	300	28	0.81	1.3%	~330 ms

From a SwiftUI `response` `T` and `dampingFraction` ζ : $\text{stiffness} = (2\pi / T)^2$, $\text{damping} = 4\pi\zeta / T$.

Timing curves

Name	iOS	Reanimated
ease-out	<code>.easeOut(duration:)</code>	<code>withTiming(v, { duration, easing: Easing.out(Easing.ease) })</code>
ease-in-out	<code>.easeInOut(duration:)</code>	<code>withTiming(v, { duration, easing: Easing.inOut(Easing.ease) })</code>
Keyboard	ease-out over the keyboard's own duration (fallback 250 ms)	<code>useAnimatedKeyboard()</code> / <code>KeyboardAvoidingView</code> , follows the system curve

React Native implementation notes

- Own the push; do not use the native stack animation.** The native iOS push has a parallax and a different curve. Either use `@react-navigation/stack` with a custom `cardStyleInterpolator` and `transitionSpec: { open: { animation: 'spring', config: P1 or P2 }, close: same }`, or render both pages in one container and drive them with Reanimated `entering` / `exiting` layout animations built from the springs above.
- Interpolator for the push.** Incoming: `translateX = (1 - progress) × W`, `opacity = progress`. Outgoing: `translateX = -progress × W`, `opacity = 1 - progress`. Both pages fade, so no shadow or dim overlay is needed.
- Set the direction first.** Decide forward or back before either page starts animating; Parent does it one frame before changing the stack so the leaving page slides the right way.
- Fixed chrome stays out of the pages.** The Parent progress bar is a layer above the navigator; only its fill animates (P2). Its colour crossfades white ↔ purple over 350 ms on Intro.
- Entrances start with the push.** Each page's entrance (Email card, Avatar column, Done, Intro) starts on mount, at the same moment as the push, with its own delay. Skin waits 350 ms so the push lands before the cards deal in.
- Timed handoffs** (fire with `setTimeout` or Reanimated callbacks):

Handoff	Delay
Splash → Email	when the video ends (~1.93 s) or on tap
OTP keypad auto-focus	680 ms after OTP mounts
OTP correct → Skin push	950 ms after the 4th digit

Handoff	Delay
Celebration → Email push	1267 ms (frame 76 of 90 at 60 fps)
Parent Email auto-focus	350 ms after mount
Intro auto-advance	1700 ms after mount
Let's Go → Invite push	on Lottie complete; unmount the Lottie 450 ms later

7. **Keyboard.** Keep inputs and action rows above the keyboard using the keyboard's own duration and curve (`react-native-keyboard-controller` or `useAnimatedKeyboard`). The Parent action bar sits 76pt above the keyboard top while it is up.
8. **Reduced motion.** Suggested, not in the iOS build: replace the push with a 200 ms crossfade and skip entrance offsets.